

example-1

Write a program that converts integer Fahrenheit temperatures from 0 to 212 degrees to floating-point Celsius temperatures with 3 digits of precision. Use the formula

```
celsius = 5.0 / 9.0 * ( fahrenheit - 32 );
```

to perform the calculation. The output should be printed in two right-justified columns of 10 characters each, and the Celsius temperatures should be preceded by a sign for both positive and negative values.

example-2

Create a file encryption processor (from decimal into binary) using `getc()`, `putc()`, `rename()`, and `stderr` error reporting.

Requirements:

- Open “source.txt” for reading (“r”) and “encrypted.txt” for writing (“w”). Validate pointers against NULL.
- Read characters using `getc(in)` until EOF: ‘0’ - ‘9’
- Write all characters (transformed and untransformed) to “encrypted.txt” using `putc(c, out)`.
- Close both files.
- Use `rename` (“encrypted.txt”, “secure_vault.txt”) to rename the output file. If `rename()` returns non-zero, write an error to `stderr` and exit with `exit (EXIT_FAILURE)`.